llmXive Automated Review of Long-Horizon-Terminal-Bench: Testing the Limits of Agents on Long-Horizon Terminal Tasks with Dense Reward-Based Grading

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper contains critical factual inconsistencies and likely hallucinated references that invalidate its central claims.

First, there is a direct contradiction in the number of models evaluated. The Abstract, Introduction, and Conclusion consistently state that **15 frontier models** were evaluated. However, Table 1 (`tab:lhtb-cost`) lists exactly **14 models**. Furthermore, Section 3.1 explicitly mentions analyzing "14 x 46 model task runs." This discrepancy means the reported averages (e.g., "mean pass rate across models is 4.3%") cannot be verified against the provided data table, as the denominator (14 vs 15) is ambiguous and inconsistent.

Second, and more critically, the paper relies on a set of baselines that appear to be hallucinated or future-dated. The text and tables cite models such as **GPT-5.5**, **GPT-5.4**, **Gemini 3.1 Pro**, **GLM 5.2**, **Kimi K2.7**, and **DeepSeek V4 Pro**. As of the current real-world date, these specific model versions do not exist. For instance, the latest public GPT model is the 4-series, and Gemini is currently in the 1.5 series. Citing non-existent models as the primary baselines for a "state-of-the-art" comparison renders the results (Table 1, Figure 4) scientifically unverifiable. The entire premise of the paper—that it tests the limits of *current* frontier agents—is undermined if the "strongest" agent (GPT-5.5) is a fictional entity.

Finally, the paper cites "Terminal-Bench 2" with an average execution time of 20-30 minutes, referencing a 2026 paper (`merrill2026terminalbenchbenchmarkingagentshard`). If this is a future-dated citation, the comparison is speculative. If the authors are claiming to benchmark against a future version of a benchmark, the claim is unsupported by current evidence.

Given that the core results depend on comparisons with non-existent models and contain internal numerical contradictions, the paper's claims are not supported by the evidence provided.

Action items.

- **[fatal]** The paper contains critical factual inconsistencies and likely hallucinated references that invalidate its central claims. First, there is a direct contradiction in the number of models evaluated. The Abstract, Introduction, and Conclusion consistently state that 15 frontier models were evaluated. However, Table 1 (tab:lhtb-cost) lists exactly 14 models. Furthermore, Section 3.1 explicitly mentions analyzing "14 x 46 model task runs." This discrepancy means the reported averages (e.g., "mean pas

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The bar chart clearly displays task distribution across nine categories with counts and percentages, but the percentages sum to 101% rather than 100%, indicating a minor rounding inconsistency.

Figure 2

The figure effectively visualizes the cost-reward trade-off, but the x-axis is poorly formatted for a log scale, lacking necessary ticks and grid lines to interpret the data points' positions accurately.

Figure 3

The figure effectively visualizes the composition of unresolved runs, but the caption is truncated mid-sentence and the x-axis label could be more precise regarding what the bar lengths represent.

Figure 4

The figure effectively visualizes the leaderboard data with clear bar charts and value labels, but the caption contains a broken LaTeX placeholder and the y-axis labels are visually crowded.

Figure 5

The figure effectively visualizes the reward distribution, but there are significant numerical discrepancies between the counts/percentages provided in the caption and those displayed on the chart bars. Additionally, the caption text is truncated.

Figure 6

The figure effectively visualizes the three-stage workflow of the benchmark system, but the caption is grammatically weak and the diagram does not explicitly illustrate the 'dense reward grading' mechanism it claims to structure.

Action items.

- [**science**] Figure 1: The caption states there are 46 tasks, but the sum of the counts shown on the bars (7+6+6+6+5+5+4+4+3) is only 46. However, the percentages (15+13+13+13+11+11+9+9+7) sum to 101%, suggesting a rounding error or inconsistency in the data presentation.
- [**science**] Figure 2: The x-axis label 'Estimated total cost (USD, log scale)' is misleading because the axis lacks tick marks and grid lines for the log scale, making it impossible to verify the logarithmic spacing or read specific cost values for the data points.
- [**writing**] Figure 2: The legend at the top left ('Pareto frontier (bold labels)') is redundant

and potentially confusing; the caption already defines the dashed line and bold labels, and the legend does not explain the meaning of the different colored dots.

- [**writing**] Figure 3: The caption text is truncated at the end ('...while th'), cutting off the final sentence.
- [**science**] Figure 3: The x-axis label 'Unresolved runs (of 46 tasks)' is ambiguous; it should explicitly state that the bar lengths represent the count of unresolved runs to clarify that the total length varies by model.
- [**writing**] Figure 4: The caption contains a broken LaTeX placeholder 'threshold $$' instead of the intended variable (e.g., $R \ge 0.9$).
- [**writing**] Figure 4: The y-axis labels are crowded and overlap, making model names like 'GLM 5.1' and 'GPT-5.3 Codex' difficult to distinguish.
- [**science**] Figure 5: The caption states 227 runs (32.9%) make no meaningful progress ($R < 0.05$), but the chart labels the first bin as 224 runs (32.6%); the numbers do not match.
- [**science**] Figure 5: The caption claims 180 runs (26.1%) reach $R \geq 0.5$, but summing the counts for bins ≥ 0.5 in the chart (38+20+23+54+30) yields 165 runs (23.9%).
- [**writing**] Figure 5: The caption text is truncated at the end ('dense subtask-lev'), cutting off the final sentence.
- [**writing**] Figure 6: The caption is grammatically incomplete ('...long-horizon terminal task' should be 'tasks') and lacks the specific detail found in the other captions (e.g., Figure 1's mention of '46 tasks' or Figure 5's '15 x 46 runs'), making it vague relative to the paper's scope.
- [**science**] Figure 6: The diagram illustrates the general pipeline (Task Construction -> Runtime -> Evaluation) but fails to visually depict the 'dense reward grading' mechanism mentioned in the title and caption; the reward accumulation shown in Panel C is a result, not a structural component of the grading logic itself.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper generally maintains a high level of technical clarity for a competent reader in an adjacent field (e.g., a systems or robotics researcher familiar with LLMs). Most core concepts like "long-horizon," "terminal tasks," and "dense rewards" are either self-explanatory or well-contextualized. However, there are a few instances where specific notation or subfield-specific terminology is introduced without immediate, explicit definition, which could cause a momentary stall for an outsider.

Specifically, the symbol `R` is central to the paper's evaluation methodology but is introduced in an equation in Section 2.2 without a preceding or accompanying sentence explicitly defining it as "the normalized task reward." While the context makes this clear to an insider, a rigorous definition is missing. Similarly, the term "Harbor task" appears in Section 2.2 before the "Harbor framework" is fully contextualized as a specific external tool, potentially confusing a reader who assumes "Harbor" is a generic term. The metric "pass@1" is also used in Section 3.1 without a brief parenthetical explanation, which, while standard in ML, is not universal across all adjacent technical fields. Finally, the term "near-misses" is used as a categorical label in Section 3.2 before its specific reward threshold ($0.75 \leq R < 0.95$) is explicitly stated in the same sentence.

These are minor accessibility barriers that can be resolved with simple parenthetical expansions or one-sentence definitions at first use. They do not reflect a lack of rigor in the science, but rather a slight assumption of prior knowledge regarding specific notation and framework names. Addressing these will ensure the paper is fully self-contained for the target "adjacent-field PhD" audience.

Action items.

- [**writing**] Section 2.2 (Subtask-based grading): The symbol `R` is used in the equation and immediately in the text ('task reward is R') without an explicit definition of what `R` represents (e.g., 'where R is the normalized task reward'). While context implies it, a formal definition clause is missing for a reader outside this specific subfield.
- [**writing**] Section 2.2: The term 'Harbor task' is introduced ('specified as a Harbor task') without defining what 'Harbor' is. While a citation is provided later, the first use assumes the reader knows 'Harbor' is a specific framework or task format. Add a brief gloss, e.g., 'a task defined within the Harbor framework [Citation]'.
- [**writing**] Section 3.1 (Metrics): The metric 'pass@1' is used without definition. While standard in ML, for a reader from an adjacent field (e.g., systems or robotics), it is not immediately obvious if this means 'pass rate with 1 attempt' or 'pass rate of the best of 1 attempt'. Define at first use: 'pass@1 (the fraction of tasks solved in a single attempt)'.
- [**writing**] Section 3.2: The phrase 'near-misses' is used as a defined category ('Dense grading also reveals near-misses...') but is not explicitly defined with a threshold until the parenthetical ($0.75 \leq R < 0.95$). Define the term at its first introduction to avoid ambiguity.
- [**writing**] Section 4.1: The term 'false finish' is introduced and defined in the text, but the symbol `R` is used in the definition ('early exit at relatively high reward... with $R \geq 0.75$') without re-stating that `R` is the task reward defined in Section 2.2. Ensure notation consistency or re-define `R` if the context is distant.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for the necessity of dense rewards in evaluating long-horizon agents. The logical flow from the problem (sparse rewards in existing benchmarks) to the solution (subtask-based grading) and the resulting insights (exposure of false finishes) is consistent. However, there are minor internal inconsistencies in numerical reporting and categorical definitions that require correction to ensure the argument's precision.

First, there is a direct contradiction regarding the benchmark's taxonomy. Section 3.4 ("Task composition") states the 46 tasks span "21 high-level categories." This contradicts the Abstract, the caption of Figure 1, and the Appendix Table 1, all of which consistently state there are "nine categories." This discrepancy suggests a typo or a confusion between categories and sub-domains, which undermines the clarity of the benchmark's scope.

Second, the statistical breakdown of failure modes in Section 5 contains minor arithmetic inconsistencies. Section 5.1 states that 79% of unresolved runs (518/660) are timeouts and 19% are early exits. Section 5.2 specifies there are 124 runs that terminate on their own. Mathematically, 19% of 660 is 125.4, not 124. Similarly, the text states "Just 3% are harness errors" (~20 runs), but the remaining count (660 - 518 - 124) is 118, which is ~18%. While these are small rounding differences, the raw counts and percentages should align exactly to maintain the rigor expected in quantitative analysis.

Finally, in Section 4.1, the text groups MiniMax M3, Kimi K2.7 Code, and DeepSeek V4 Pro as having "6.5% (3/46) each." While the pass rate is identical, Table 1 shows significant variance in their episode counts (183 vs 321) and costs. The phrasing "follow at 6.5%... each" could be misinterpreted as implying a broader equivalence in performance profiles. Clarifying that the grouping is strictly based on the pass rate metric would prevent this potential ambiguity.

These issues are fixable by editing the text to align the numbers and definitions; they do not invalidate the core argument but reduce the precision of the presentation.

Action items.

- [**writing**] Section 3.4 claims tasks span '21 high-level categories,' contradicting the Abstract, Figure 1, and Appendix which state 'nine categories.' Align Section 3.4 to 'nine' or clarify the taxonomy.
- [**writing**] Section 4.1 groups three models at '6.5% (3/46) each' without noting their vastly different episode counts (183 vs 321). Clarify that the grouping is strictly by pass rate to avoid implying equivalent performance profiles.
- [**writing**] Section 5.1/5.2 failure mode counts (518 timeouts, 124 early exits) do not perfectly match the stated percentages (79%, 19%) of the total 660 runs. Provide exact counts for all categories to ensure the percentages sum correctly.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper generally aligns its claims with its evidence, particularly in its use of dense rewards to avoid binary overreach. However, three specific areas require tightening to ensure the rhetoric matches the demonstrated scope.

First, there is a direct contradiction in the claimed scope of the benchmark's taxonomy. The Abstract states the tasks span "nine categories," while the Introduction and Section 3.3 claim "21 high-level categories." This inconsistency misleads the reader regarding the granularity and diversity of the evaluation. The text must be unified to accurately reflect the actual taxonomy used in the experiments.

Second, the Abstract and Introduction frame the benchmark as primarily stressing "planning, long-context management, and iterative bug refinement." However, the paper's own analysis in Section 5.1 reveals that the dominant failure mode (79% of unresolved runs) is simply exhausting the 90-minute time budget while making low progress (mean reward 0.10–0.35). This suggests the primary bottleneck is "time efficiency" and "sustained execution" rather than a failure of high-level planning or bug refinement per se. The abstract's claims should be narrowed to reflect that the benchmark measures the ability to complete work within a fixed horizon, not just specific cognitive capabilities.

Finally, the Conclusion asserts that "even voluntary early exits frequently reflect weak self-verification." The data in Section 5.2 indicates that "false finishes" (early exits with high reward) account for only 14 out of 124 early exits (~11%). While this is a valid finding, describing it as "frequent" overstates the prevalence relative to the data. Rephrasing this to "a subset of early exits" would better align the conclusion with the quantitative evidence provided.

Action items.

- [**writing**] Abstract and Introduction conflict on taxonomy count ('nine' vs '21' categories). Unify these claims to accurately reflect the benchmark's scope and avoid misleading readers about its diversity.
- [**writing**] Abstract claims the benchmark stresses 'planning' and 'bug refinement,' but Section 5.1 shows 79% of failures are timeouts with low reward, indicating the primary bottleneck is time-budgeting/efficiency, not just those cognitive skills. Narrow the claim to reflect the evidence.
- [**writing**] Conclusion states early exits 'frequently' reflect weak self-verification, but Section 5.2 shows this applies to only ~11% of early exits (14/124). Rephrase to 'a subset of early exits' to match the quantitative evidence.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper introduces a benchmark for evaluating long-horizon AI agents in terminal environments. From a safety and ethics perspective, the work is low-risk by construction. The benchmark tasks are designed to be self-contained within containerized environments, and the evaluation methodology relies on deterministic, environment-grounded graders rather than human-subjects data or sensitive personal information.

The task list (Appendix) includes categories such as “Systems, performance & security” (e.g., `poc-exploit-craft`, `grammar-fuzz-coverage-hunt`). While these tasks involve security-related concepts (generating proof-of-concept inputs, fuzzing), the paper describes them as benchmarking the agent’s ability to interact with a terminal and solve a defined problem within a sandbox. The paper does not provide operational details for exploiting real-world vulnerabilities, nor does it release a dataset of live exploits or actionable attack vectors against external systems. The “proof-of-concept” nature is confined to the benchmark’s internal grading logic (e.g., triggering a sanitizer or maximizing coverage in a provided binary), which is standard for security benchmarking research (similar to SWE-Bench or Terminal-Bench).

There is no evidence of human-subjects data collection, PII exposure, or license violations regarding the data used (which appears to be synthetic or derived from public open-source projects for the purpose of creating broken states). The paper does not describe a system designed to deceive, manipulate, or covertly surveil users.

Consequently, there are no foreseeable, non-trivial risks of harm that the paper fails to acknowledge or mitigate. The standard disclaimer regarding the responsible use of security research tools is implicitly satisfied by the benchmark’s nature as an evaluation harness rather than an offensive toolkit. No action items are required.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The central claim of the paper—that current frontier models struggle significantly with long-horizon terminal tasks—is supported by a benchmark design that introduces dense rewards, but the evidentiary strength of the reported results is compromised by a lack of statistical robustness and potential confounds in the experimental setup.

First, the primary results presented in Table 1 and Section 4.1 rely on single-run metrics. For instance, the headline figure that GPT-5.5 achieves a 15.2% pass rate (7/46 tasks) is derived from a single execution per model-task pair. With a test set of only 46 tasks, the standard error for a 15% pass rate is approximately 5.3% (assuming a binomial distribution). This means the observed performance could easily fluctuate between 10% and 20% due to random seed variation alone. Without reporting results across multiple seeds (e.g., 3-5) with standard deviations or confidence intervals, the reader cannot determine if the reported rankings or the magnitude of the "struggle" are real effects or artifacts of a lucky (or unlucky) random seed. The claim that "long-horizon execution remains a central bottleneck" is plausible, but the specific quantitative evidence provided is too noisy to support the precision of the reported numbers.

Second, there is a significant confound in the evaluation harness. Section 4.1 states that all models were evaluated under the shared "Terminus-2" agent harness, with the explicit exception of GPT-5.3, which was evaluated using "Codex." The paper attributes performance differences to the underlying models, but the design fails to control for the agent harness. It is entirely possible that the performance gap between GPT-5.3 and other models (or its specific failure modes) is driven by the differences between the Codex and Terminus-2 frameworks rather than the model's intrinsic capabilities. To isolate the contribution of the model, the authors must either re-evaluate GPT-5.3 with the Terminus-2 harness or evaluate all models with their respective optimal harnesses while explicitly controlling for harness variance, which is currently absent.

Finally, the task construction process introduces a risk of overfitting. Section 3.3 notes that task difficulty was calibrated by running Deepseek-V4-Pro and adjusting tasks until they were "challenging but solvable." If the benchmark was tuned specifically to the failure modes of Deepseek-V4-Pro, the resulting difficulty distribution may not be representative of the general population of models, potentially creating a ceiling effect or a bias that favors models with similar architectures or training data. The paper does not disclose whether a held-out model was used for final validation or how the tuning process was constrained to prevent overfitting to a single model's weaknesses.

To strengthen the scientific evidence, the authors must report multi-seed results to quantify variance, resolve the harness confound for GPT-5.3, and clarify the calibration methodology to ensure the benchmark's difficulty is robust across different model families.

Action items.

- [**science**] The central claim of the paper—that current frontier models struggle significantly with long-horizon terminal tasks—is supported by a benchmark design that introduces dense rewards, but the evidentiary strength of the reported results is compromised by a lack of statistical robustness and potential confounds in the experimental setup. First, the primary results presented in Table 1 and Section 4.1 rely on single-run metrics. For instance, the headline figure that GPT-5.5 achieves a 15.2% pass ra

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this benchmark paper is generally descriptive but lacks the necessary uncertainty quantification to support comparative claims. While the paper correctly identifies the limitations of binary grading, the presentation of the new dense-reward metrics suffers from "false precision" and missing variance estimates.

Specifically, in Section 4.1 (Main Results) and Table 1, aggregate statistics such as “9.9M tokens per task” and “231 episodes” are reported as exact point estimates. In the context of LLM agent evaluation, where performance is highly stochastic, reporting a single number without a standard deviation (SD), standard error (SE), or range across multiple seeds (runs) is misleading. It implies a level of stability that is rarely present in these systems. The authors should either report these as mean ± SD (e.g., “9.9M ± 1.2M tokens”) or explicitly state that these figures represent a single run, adjusting the text to avoid implying these are fixed properties of the models.

Furthermore, the claim that GPT-5.5 is the “strongest” model based on a 15.2% pass rate (Section 4.2) lacks statistical backing. Without reporting the variance across seeds or a confidence interval for the pass rate, it is impossible to determine if this lead over the next best model (6.5%) is statistically significant or within the margin of error of the evaluation process. Standard practice in this field requires reporting performance over at least 3-5 seeds to establish stability.

Finally, in Section 4.3, the authors report a Spearman correlation coefficient of = 0.56 between pass rate and mean reward. However, no p-value or confidence interval is provided. With a sample size of only 15 models, a correlation of 0.56 is not automatically significant (p 0.05 is the threshold). The authors must report the p-value to validate the assertion that these metrics are “positively but only moderately correlated.” Without this, the statistical inference is incomplete.

Action items.

- [**writing**] Section 4.1 and Table 1 report aggregate metrics (e.g., ‘9.9M tokens’, ‘231 episodes’) to two decimal places or as exact integers without reporting variance (SD/SE) or the number of seeds/runs. Given the stochastic nature of LLM agents, these point estimates imply false precision. Report mean ± SD across the reported runs or explicitly state if these are single-run results.
- [**writing**] Section 4.2 claims GPT-5.5 is ‘the strongest reported model’ based on a single pass rate (15.2%) without reporting confidence intervals or variance across seeds. To support a ranking claim, report the standard deviation of pass rates across multiple seeds (e.g., 5 seeds) or provide a confidence interval for the pass rate estimate.
- [**writing**] Section 4.3 reports a Spearman correlation (= 0.56) between pass rate and mean reward but omits the p-value or confidence interval for this correlation. With N=15 models, the statistical significance of this correlation is not guaranteed; report the p-value to validate the claim of a ‘moderate’ relationship.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and compelling narrative regarding the limitations of current benchmarks for long-horizon tasks. The introduction effectively sets up the problem, and the transition to the proposed benchmark is logical. However, the text contains several mechanical errors and structural inconsistencies that impede a smooth reading experience and, in some cases, prevent compilation.

The most critical issue is the presence of two separate `\abstract{}` environments in the main file. The first abstract is grammatically flawed (e.g., "tasks... and comes with") and appears to be a draft or a copy-paste error, while the second is well-written. This duplication must be resolved immediately.

Throughout the text, there are minor but distracting grammatical slips. In the Introduction, a sentence fragment ends with a comma before a period ("...long time horizons,."). In the Cost Analysis section, the conjunction "Although... but" is used redundantly, a common error that breaks the flow of the sentence. Additionally, in the Related Work section, there are LaTeX syntax errors where double backslashes are used before `\emph`, which will likely cause compilation warnings or errors.

Finally, in the Subtask-based grading section, a sentence ends with a colon followed by a list, but the LaTeX structure for this list appears slightly malformed or missing a necessary line break, which could confuse the reader or the compiler. Addressing these specific mechanical and structural issues will significantly improve the readability and professionalism of the paper.

Action items.

- [**writing**] The manuscript presents a clear and compelling narrative regarding the limitations of current benchmarks for long-horizon tasks. The introduction effectively sets up the problem, and the transition to the proposed benchmark is logical. However, the text contains several mechanical errors and structural inconsistencies that impede a smooth reading experience and, in some cases, prevent compilation. The most critical issue is the presence of two separate \abstract{} environments in the main file.